

TECHNICAL BRIEFING SUMMARY: DISTRIBUTED SIEM/XDR INFRASTRUCTURE & QUANTITATIVE RISK AUTOMATION

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Target Position: Cyber Analyst, Junior (Req. R0249110)

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1. EXECUTIVE SUMMARY & OBJECTIVE

The objective of this technical initiative was to architect and deploy a self-contained **Security Information and Event Management (SIEM)** and **Extended Detection and Response (XDR)** telemetry pipeline within a localized sandbox environment. Built entirely upon open-source enterprise frameworks, this production architecture functions as an automated data engine that ingests, cleans, and validates raw endpoint telemetry logs, translating qualitative indicators into structured, baseline risk-compliance datasets to establish consistent corporate audit readiness.

2. CORE ARCHITECTURAL FLOW & APPLICABLE PROJECTS

Enterprise Cybersecurity & Threat Hunting Engine | Independent Portfolio Project

- **Log Ingestion & Parsing:** Engineered and tuned an operational **Wazuh SIEM/XDR pipeline** within an isolated hypervisor sandbox running a multi-OS configuration. The central manager ingests unstructured system log streams (including Windows Event Logs and Linux auth.log), normalizing disjointed signatures into standardized telemetry metrics.
- **Distributed Endpoint Coverage:** Deployed lightweight monitoring agents securely across target sandboxes (including Windows 11 and Ubuntu virtual nodes) to forward system events, process states, and event IDs to the manager over encrypted standard ports.
- **MITRE ATT&CK; Mapping:** Investigated active security alerts and correlated indicators of compromise (IOCs) to map anomalous behavioral patterns directly against the **MITRE ATT&CK; framework Matrix** for real-time risk visibility.
- **Quantitative Risk Automation:** Programmed a custom Python-based mathematical simulator executing **10,000+ Monte Carlo iterations**. This script parses live alert counts from the log ingestion database and calculates definitive value-at-risk matrices for enterprise assets, converting complex alert telemetry into concise financial and operational exposure narratives for senior leadership.
- **Visualizing Results:** Built a secure, public-facing web portfolio hosted on a personal GitHub repository (/spardman) using clean, optimized code to dynamically visualize live system telemetry and threat data feeds for stakeholders.

Integrated Consulting Simulations | Boston Consulting Group (BCG) / Oliver Wyman

- **Insight Generation:** Dissected and consolidated complex technical data streams for global enterprise clients to diagnose system-wide performance gaps, technical vulnerabilities, and operational infrastructure risks.
- **Executive Narratives:** Partnered with cross-functional IT teams to design executive-facing dashboards, successfully translating complex technical findings into actionable risk-mitigation roadmaps for senior leadership.

3. OPERATIONS & DATA GOVERNANCE TRACKING

The functional methodologies deployed in these projects leverage a career baseline of managing data integrity and platform configurations across public sector and corporate infrastructures scaling up to **30,000 active stakeholders**:

- **Least Privilege Enforcement:** Audited user access boundaries and identity management logs to enforce robust data defenses and logical perimeter access.
- **Framework Alignment:** Mapped observed lab event flows directly to federal and regulatory compliance frameworks—including **NIST SP 800-53, ITAR, and FERPA**—to maintain absolute record integrity and data privacy controls.
- **Actionable SOP Development:** Formulated clear, comprehensive Standard Operating Procedures (SOPs) and technical documentation tailored to bridge the gap between technical infrastructure operations and high-level corporate risk governance.

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